# AI-Powered Clustering Insights

## Project Overview:

In many organizations, raw data just sits in spreadsheets without being used to make decisions. Business teams often struggle to figure out which customer segments, product groups, or behavior patterns matter most — and analysts waste hours manually testing clustering settings. We wanted to solve this by creating an AI-powered clustering assistant that reads a dataset, recommends the best clustering setup, runs the analysis, and gives clear, easy-to-read insights.

## Why Needed?

- **Time saving:** Cuts down hours of trial-and-error clustering by recommending the most promising configuration instantly.
- **Better decisions:** Provides high-level summaries that managers can understand, plus detailed metrics for analysts.
- **No deep data science skills required:** Works for users who can upload a dataset and click through a simple UI.
- **Consistent results:** AI chooses clustering parameters based on actual data patterns, not guesswork.

## Technology Stack:

- **Framework/UI:** Streamlit
- **Data Processing:** pandas, numpy
- **Clustering Algorithms:** scikit-learn (KMeans, Agglomerative)
- **Visualization:** plotly, seaborn, matplotlib
- **AI Recommendations:** Google Gemini & Groq API integration
- **Other Tools:** requests (API calls), pathlib (file handling)

## Current Capabilities:

- Upload CSV datasets (with automatic quality checks).
- User reviews recommendation, clicks Run to execute clustering.
- After clustering, user clicks Generate Insights to get:
- Business Insights
- Cluster Metrics Table (top-level metrics only)
- Interactive Dashboards (distribution plots, cluster comparison charts)
- Results available on-screen and for instant download.

## Workflow:

1. Data Upload – User uploads a CSV file.
2. Data Validation – App checks for missing values, duplicates, and identifies numeric columns.
3. Feature Selection – User chooses numeric features to include.
4. On selected features apply algorithm.
5. Run Clustering – User confirms and runs clustering.
6. Insights Generation
7. Visual dashboards
8. Business insights by AI
9. Download – User downloads full results (CSV & visuals) if needed.

## Target Users:

- **Business Managers:** Get quick, understandable summaries without diving into raw stats.
- **Data Analysts:** Save time by starting from AI's optimal configuration instead of manual trial-and-error.
- **Students & Researchers:** Quick prototyping tool for clustering experiments

## Next Phase:

- AI recommends:
  - Best clustering algorithm (e.g., KMeans vs Agglomerative)
  - Optimal K value (number of clusters)
  - Pros/cons of the choice

- Improve AI's reasoning for recommendations & add more clustering algorithms (e.g., HDBSCAN).